Cairo University
Faculty of Engineering
Electronics and Electrical Communications Engineering

Assignment 1

ELC 4028 - Artificial Neural Networks

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PART1 : Regressing Problems

Problem 1: Insurance Company

The numbers of insured persons y with an insurance company for the years 1987 to 1996 are given. We let x represent the number of years since 1987 (x = 0 for 1987, x = 9 for 1996).

1.1 Dataset

x (years since 1987)	Year	y (insured persons)
0	1987	12,400
1	1988	10,900
2	1989	10,000
3	1990	**1,050 (OUTLIER)**
4	1991	9,500
5	1992	8,900
6	1993	8,000
7	1994	7,800
8	1995	7,600
9	1996	7,200

** Value 1050 at x = 3 (year 1990) is the outlier — likely a data entry error or anomaly.*

1.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification A scatter plot of y (number of insured persons) versus x (years since 1987) reveals a generally decreasing trend from ~12,400 down to ~7,200. However, one data point stands out sharply:

- x = 3 (year 1990): y = 1050

This value is dramatically lower than the surrounding data , which are all within the range of approximately **7,200 to 12,400**. The sharp drop at this point breaks the otherwise smooth downward trend.

Therefore, this observation is identified as a clear outlier, most likely due to a data entry error (for example, a missing digit such as *10,500* instead of *1,050*).

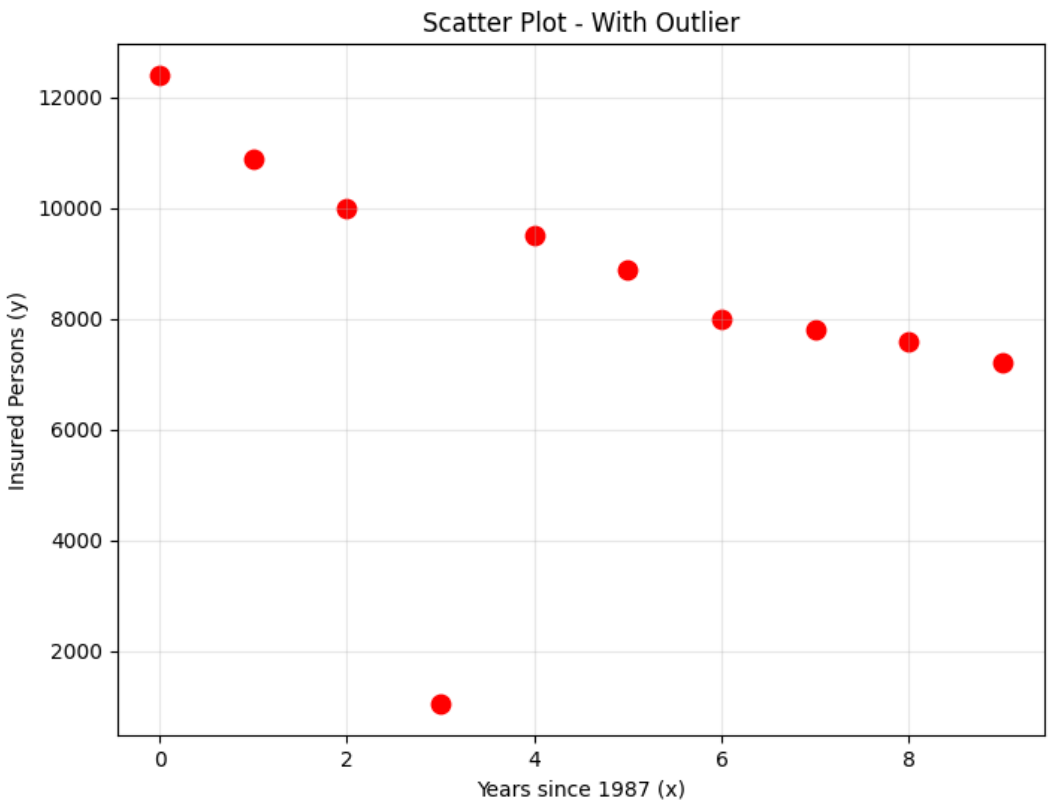

Figure 1: Scatter plot of original dataset identifying the outlier at x = 3 (y = 1050)
dataset identifying the outlier at x = 3 (y = 1050)

2. Regression Models (WITH Outlier) We apply both OLS and matrix-form methods to compute linear model parameters, then use the matrix form for quadratic and cubic models.

2.1 Linear Model

•

The equation for the linear model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i$$

The key formulas used are:

$$\widehat{\beta_1} = \frac{\sum_{i=i}^{n} x_i y_i \; - \; \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 \; - \; \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \; , \; \widehat{\beta_0} = \widehat{y} \; - \; \widehat{\beta_i} x$$

Data Table for Hand Computation:

year	$\mathbf{x}_i$	$\mathbf{y}_i$	${\mathbf{x}_i}^2$	${\mathbf{y}_i}^2$	$\mathbf{x}_i\mathbf{y}_i$
1987	0	12400	0	153,760,000	0
1988	1	10900	1	118,810,000	10,900
1989	2	10000	4	100,000,000	20,000
1990	3	1050	9	1,102,500	3,150
1991	4	9500	16	90,250,000	38,000
1992	5	8900	25	79,210,000	44,500
1993	6	8000	36	64,000,000	48,000
1994	7	7800	49	60,840,000	54,600

year	$\mathbf{x}_i$	$\mathbf{y}_i$	${\mathbf{x}_i}^2$	${\mathbf{y}_i}^2$	$\mathbf{x}_i\mathbf{y}_i$
1995	8	7600	64	57,760,000	60,800
1996	9	7200	81	51,840,000	64,800
$\Sigma =$	**45**	**83,350**	**285**	**777,572,500**	**344,750**

$n \ = \ 10$, $y = \frac{83350}{10} = 8335$, $x = \frac{45}{10} = 4.5$

$\widehat{\beta_1} = \frac{344{,}750 \ - \ \frac{45 \ * \ 83{,}350}{10}}{285 - \ \frac{45^2}{10}} \ = \ - \ 367.57$

$$\widehat{\beta_0} = 8335 \ + 367.57 \ * \ \ 4.5 = 9989.09$$

•

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \ , \ \ X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix} \ , \ Y \ = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 9989.09 \\ -367.576 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = \ 9989.09 - 367.576 \ x$$

•

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_\mathbf{0} + \beta_\mathbf{1}\mathbf{x_i}$	$\mathbf{SSE} = (\mathbf{y}_i - \hat{y}_i)^2$	$\mathbf{SSyy} = (\mathbf{y}_i - \bar{y})^2$
1987	0	12400	9,989.1	5,812,483	16,524,225
1988	1	10900	9,621.5	1,634,524	6,579,225
1989	2	10000	9,253.9	556,606	2,772,225
1990	3	1050	8,886.4	61,408,600	53,071,225
1991	4	9500	8,518.8	962,777	1,357,225
1992	5	8900	8,151.2	560,683	319,225
1993	6	8000	7,783.6	46,813	112,225
1994	7	7800	7,416.1	147,410	286,225
1995	8	7600	7,048.5	304,169	540,225
1996	9	7200	6,680.9	269,455	1,288,225
$\Sigma =$	**45**	**83,350**	**83,350**	**71,703,515**	**82,850,250**

$$R^2_{linear\ model} = \frac{SS - SSE}{SS} = \frac{82,850,250 - 71,703,515}{82,850,250} = 0.13454$$

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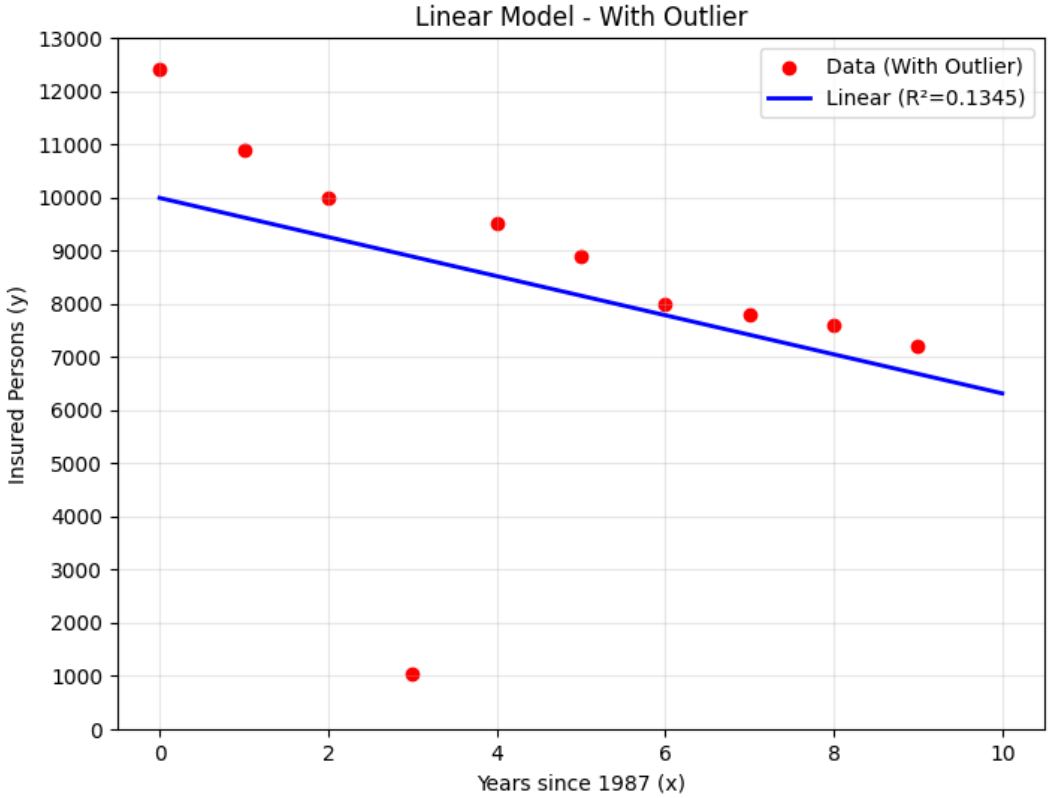

Figure 2: Linear Model of Original Data with Outlier

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Linear Model:
  Equation: y = 9989.09 + (-367.58)*x^1
  R^2: 0.1345
```

Figure 3: Simulation Results of Linear Model of Original Data with Outlier

2.2 Quadratic Model The equation for the quadratic model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

•

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \quad , \quad X = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix} \quad , \; Y \; = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 11627.73 \\ -1596.55 \\ 136.55 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = 11627.73 - 1596.55x + 136.55x^2$$

•

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_0 + \beta_1 \mathbf{x_i} + \beta_2 \mathbf{x_i^2}$	**SSE** $= (\mathbf{y}_i - \hat{y}_i)^2$	**SSyy** $= (\mathbf{y}_i - \bar{y})^2$
1987	0	12400	**11,627.7**	**596,405**	**16,524,225**
1988	1	10900	10,167.7	536,223	6,579,225
1989	2	10000	8,980.8	1,038,701	2,772,225
1990	3	1050	8,067.0	49,238,925	53,071,225
1991	4	9500	7,426.4	4,299,969	1,357,225
1992	5	8900	7,058.8	3,390,064	319,225
1993	6	8000	6,964.3	1,072,639	112,225
1994	7	7800	7,143.0	431,710	286,225
1995	8	7600	7,594.7	28	540,225
1996	9	7200	8,319.5	1,253,377	1,288,225
$\Sigma =$	**45**	**83,350**	83,350	61,858,042	82,850,250

$$R^2_{quad\ model} = \frac{SS - SSE}{SS} = \frac{82,850,250 - \mathbf{61,858,042}}{82,850,250} = 0.2534$$

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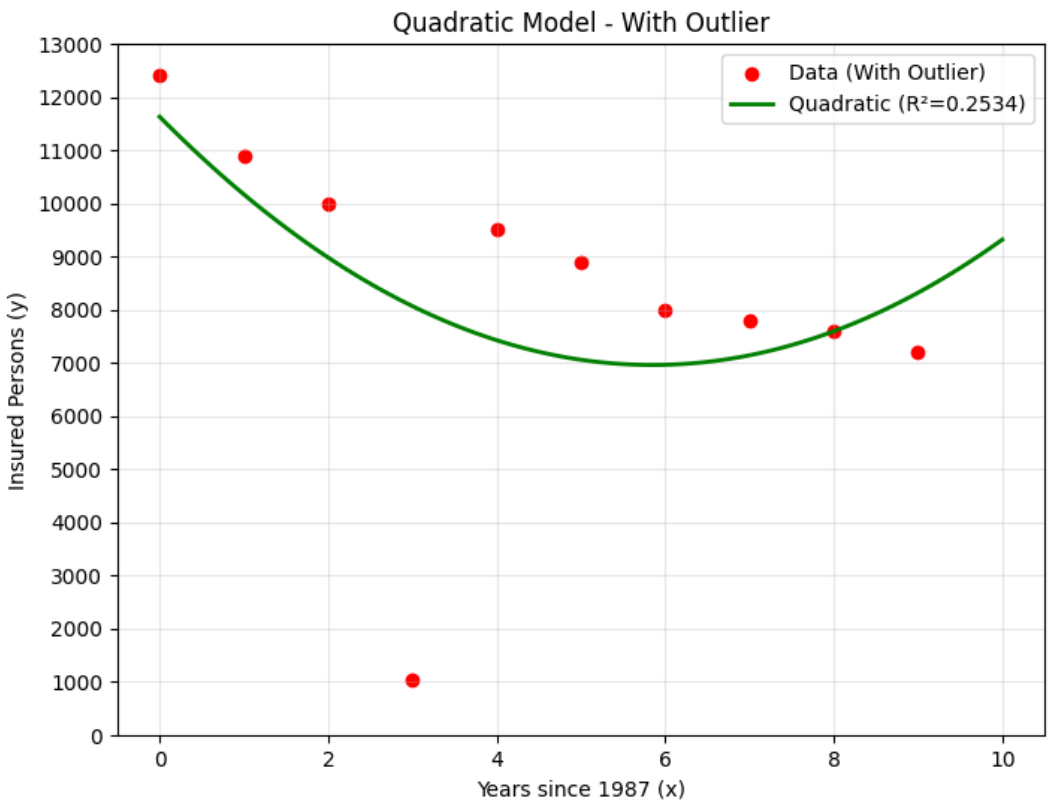

Figure 4: Quadratic Model of Original Data with Outlier

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Quadratic Model:
  Equation: y = 11627.73 + (-1596.55)*x^1 + (136.55)*x^2
  R^2: 0.2534
```

Figure 5: Simulation Results of Quadratic Model of Original Data with Outlier

2.3 Cubic Model The equation for the cubic model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

•

$$\beta = (\mathbf{X^T X})^{-1} \mathbf{X^T Y}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0^2 & 0^3 \\ 1 & 1^2 & 1^3 \\ 1 & 2^2 & 2^3 \\ 1 & 3^2 & 3^3 \\ 1 & 4^2 & 4^3 \\ 1 & 5^2 & 5^3 \\ 1 & 6^2 & 6^3 \\ 1 & 7^2 & 7^3 \\ 1 & 8^2 & 8^3 \\ 1 & 9^2 & 9^3 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 13113.15 \\ -4313.93 \\ 932.31 \\ -58.95 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = 13113.15 - 4313.93\ x_i\ +\ 932.31\ x_i^2\ -\ \ 58.95\ x_i^3$$

•

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_0 + \beta_1 \mathbf{x_i} + \beta_2 \mathbf{x_i^2} + \beta_3 \mathbf{x_i^3}$	**SSE** $= (\mathbf{y}_i - \hat{y}_i)^2$	**SSyy** $= (\mathbf{y}_i - \bar{y})^2$
1987	0	12400	**13,113.1**	**508,578**	**16,524,225**
1988	1	10900	9,672.6	1,506,542	6,579,225
1989	2	10000	7,743.0	5,094,123	2,772,225
1990	3	1050	6,970.7	35,054,266	53,071,225
1991	4	9500	7,002.0	6,240,214	1,357,225
1992	5	8900	7,483.2	2,007,341	319,225
1993	6	8000	8,060.7	3,684	112,225
1994	7	7800	8,380.8	337,334	286,225
1995	8	7600	8,089.8	239,940	540,225
1996	9	7200	6,834.1	133,864	1,288,225
$\Sigma =$	**45**	**83,350**	83,350	51,125,885	82,850,250

$$R^2_{cubic\ model} = \frac{SS\ -\ SSE}{SS} = \frac{82,850,250\ \mathbf{-51,125,885}}{82,850,250} = 0.3829$$

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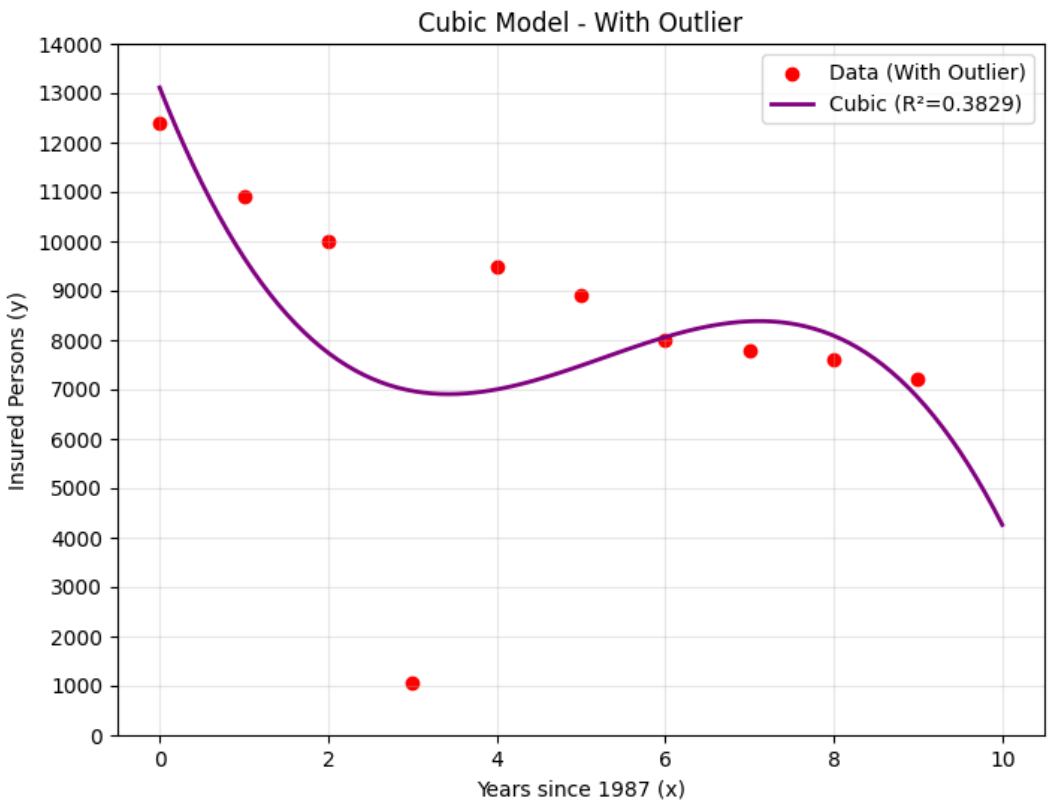

Figure 6: Cubic Model of Original Data with Outlier

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Cubic Model:
  Equation: y = 13113.15 + (-4313.93)*x^1 + (932.31)*x^2 + (-58.95)*x^3
  R^2: 0.3829
```

Figure 7: Simulation Results of Cubic Model of Original Data with Outlier

3. Comparison between three models (WITH Outlier):

Model	Equation (MATLAB)	R^2
Linear	$\widehat{y}_i = 9989.09 - 367.576\ x_i$	0.1345
Quadratic	$\widehat{y}_i = 11627.73 - 1596.55\ x_i + 136.55\ x_i^2$	0.2534
Cubic	$\widehat{y}_i = 13113.15 - 4313.93\ x_i + 932.31\ x_i^2 - 58.95\ x_i^3$	0.3829

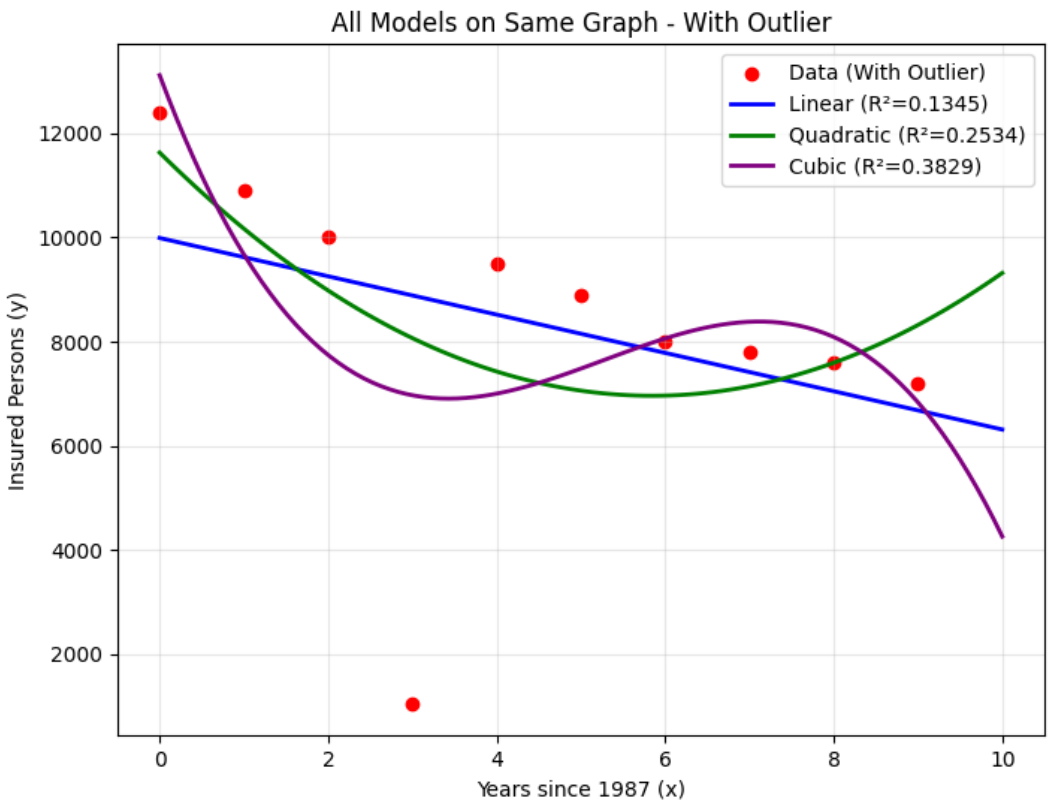

Figure 8: Comparison between three Models of Original Data with Outlier

Comment:

All R^2 values are low (< 0.50) due to the heavy influence of the outlier at x = 3 (y = 1050). The cubic model yields the highest R.

1.3 Case B: WITHOUT Outlier

1. Outlier Removal and Scatter Plot of Cleaned Data

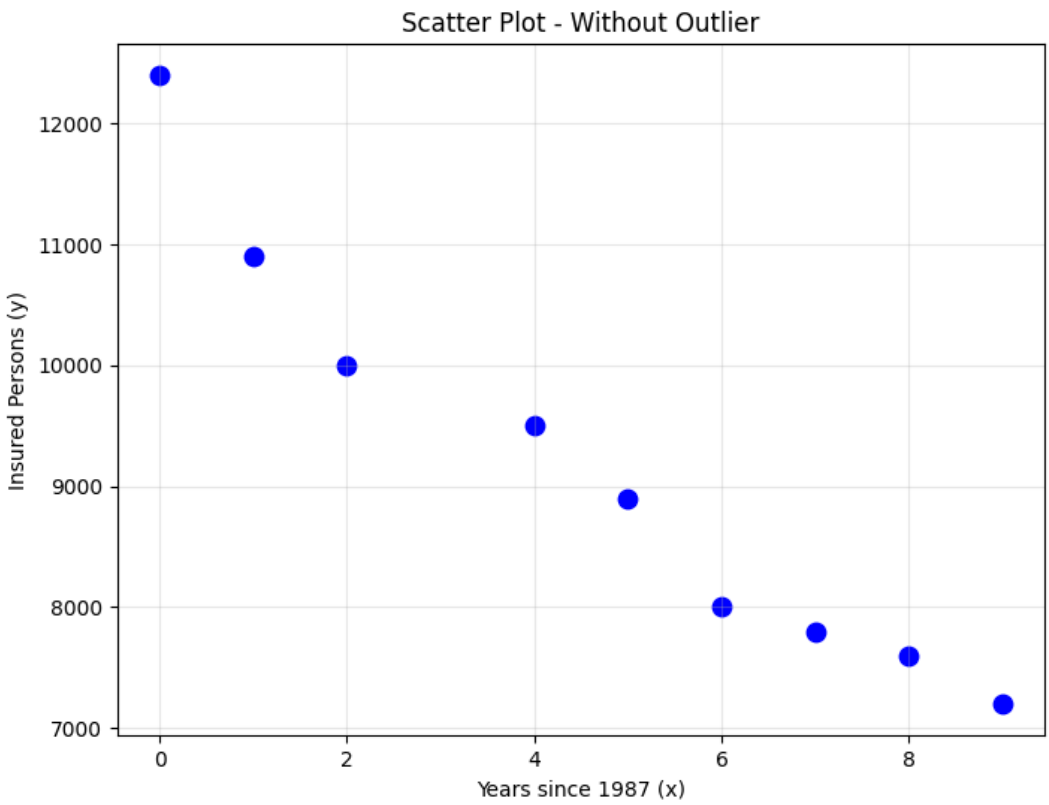

Figure 9: Scatter Plot of Cleaned Data without Outlier

2. Regression Models (WITHOUT Outlier) After removing the outlier at x = 3 (y = 1,050), the cleaned dataset contains 9 data points. We now repeat the regression analysis using the same methods.

2.1 Linear Model

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The equation for the linear model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i$$

Data Table for Hand Computation:

year	$\mathbf{x}_i$	$\mathbf{y}_i$	${\mathbf{x}_i}^2$	${\mathbf{y}_i}^2$	$\mathbf{x}_i\mathbf{y}_i$
1987	0	12,400	0	153,760,000	0
1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
$\Sigma =$	**42**	**82,300**	**276**	**776,470,000**	**341,600**

$n = 9$, $y = \frac{82300}{9} = 9,144.44$, $x = \frac{42}{9} = 4.667$

$\widehat{\beta_1} = \frac{341{,}600 \ - \ \frac{42*82300}{9}}{276 - \ \frac{42^2}{9}} \ = \ - \ 530.83$

$$\widehat{\beta_0} = 9,144.44 \ + 530.83 \ * \ 4.667 = 11,621.67$$

•

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \ , \ X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix} \ , \ Y \ = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 11,621.67 \\ -530.83 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = 11,621.67 \ - 530.83 \ x$$

•

$$\widehat{y_i} = \frac{82300}{\mathbf{9}} = 9,144.44$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_0 + \beta_1 \mathbf{x_i}$	$\mathbf{SSE} = (\mathbf{y}_i - \hat{y}_i)^2$	$\mathbf{SSyy} = (\mathbf{y}_i - \bar{y})^2$
1987	0	12400	11621.7	605,803	10,598,642
1988	1	10900	11090.8	36,417	3,081,975
1989	2	10000	10560.0	313,600	731,975
1991	4	9500	9498.3	3	126,420
1992	5	8900	8967.5	4,556	59,753
1993	6	8000	8436.7	190,678	1,309,753
1994	7	7800	7905.8	11,201	1,807,531
1995	8	7600	7375.0	50,625	2,385,309
1996	9	7200	6844.2	126,617	3,780,864
$\Sigma =$	**42**	**82,300**	**82300.0**	**1,339,500**	**23,882,222**

$$R^2_{linear\ model} \ = \ \frac{SS \ - \ SSE}{SS} \ = \ \frac{23,882,222 \ - \ 1,339,500}{23,882,222} \ = \ 0.9439$$

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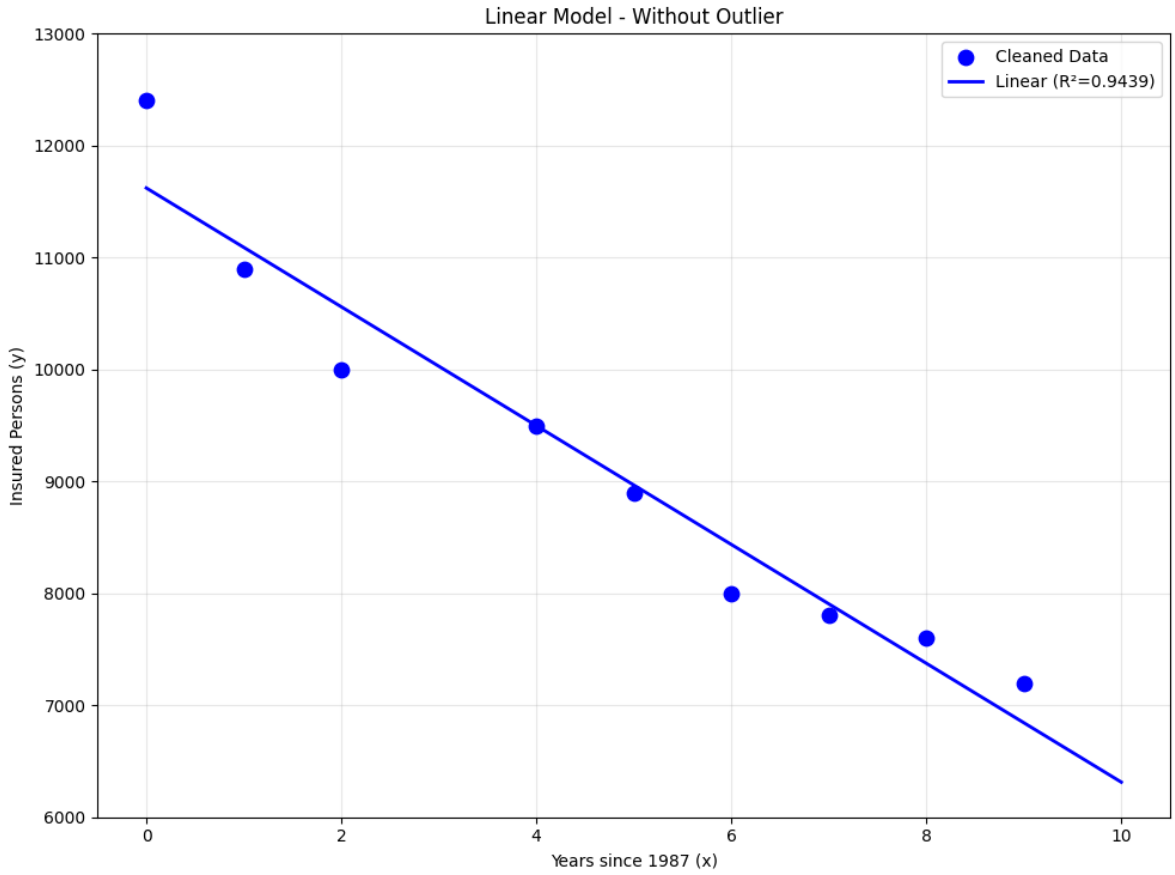

Figure 10: Linear Model of Cleaned Data without Outlier

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Linear Model:
  Equation: y = 11621.67 + (-530.83)*x^1
  R^2: 0.9439
```

Figure 11: Simulation Results of Linear Model of Cleaned Data without Outlier

2.2 Quadratic Model The equation for the quadratic model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

•

$$\beta = (\mathbf{X^T X})^{-1} \mathbf{X^T Y}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \quad , \quad \mathbf{X} = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix} \quad , \quad \mathbf{Y} = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \mathbf{12,024.17} \\ \mathbf{-863.13} \\ \mathbf{37.44} \end{pmatrix}$$

So ,linear equation is:

$$\widehat{\mathbf{y_i}} = \mathbf{12,024.17} \ - \mathbf{863.13\ x} + \mathbf{37.44\ x^2}$$

•

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{\mathbf{82,300}}{9} = 9,144.44$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_0 + \beta_1 \mathbf{x_i} + \beta_2 \mathbf{x_i^2}$	$\mathbf{SSE} = (\mathbf{y}_i - \hat{y}_i)^2$	$\mathbf{SSyy} = (\mathbf{y}_i - \bar{y})^2$
1987	0	12400	12024.2	141,248	10,598,642
1988	1	10900	11198.5	89,090	3,081,975
1989	2	10000	10447.7	200,411	731,975
1991	4	9500	9170.7	108,429	126,420
1992	5	8900	8644.6	65,249	59,753
1993	6	8000	8193.3	37,362	1,309,753
1994	7	7800	7816.9	286	1,807,531
1995	8	7600	7515.4	7,156	2,385,309
1996	9	7200	7288.8	7,884	3,780,864
$\Sigma =$	**42**	**82,300**	**82300.0**	**657,116**	**23,882,222**

$$\mathbf{R^2_{quad\ model}} = \frac{\mathbf{SS\ -\ SSE}}{\mathbf{SS}} = \frac{\mathbf{23,882,222 - 657,116}}{\mathbf{23,882,222}} = \mathbf{0.9725}$$

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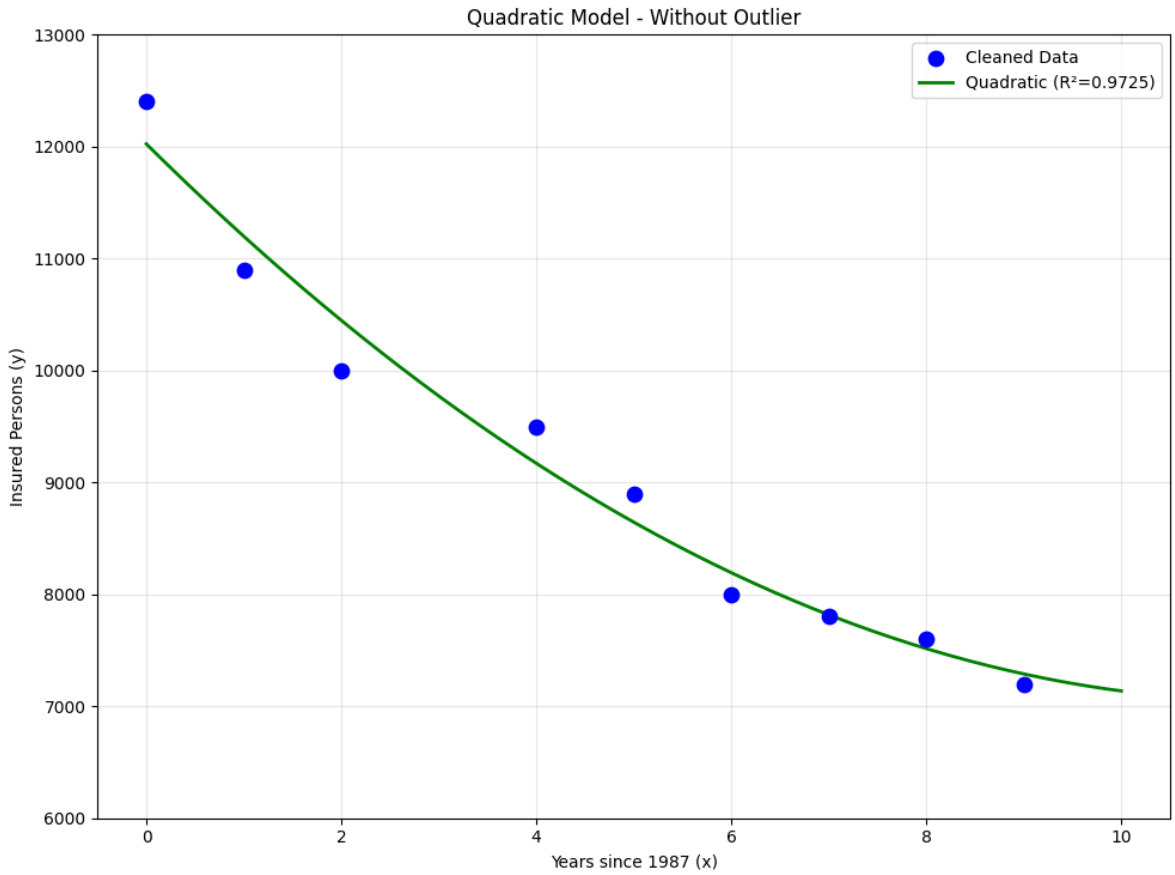

Figure 12: Quadratic Model of Cleaned Data without Outlier

Cleaned Data without Outlier

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Quadratic Model:
  Equation: y = 12024.17 + (-863.13)*x^1 + (37.44)*x^2
  R^2: 0.9725
```

Figure 13: Simulation Results of Quadratic Model of Cleaned Data without Outlier

2.3 Cubic Model The equation for the cubic model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

•

$$\beta = (\mathbf{X^T X})^{-1} \mathbf{X^T Y}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0^2 & 0^3 \\ 1 & 1^2 & 1^3 \\ 1 & 2^2 & 2^3 \\ 1 & 4^2 & 4^3 \\ 1 & 5^2 & 5^3 \\ 1 & 6^2 & 6^3 \\ 1 & 7^2 & 7^3 \\ 1 & 8^2 & 8^3 \\ 1 & 9^2 & 9^3 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 12,204.42 \\ -1,221.74 \\ 140.17 \\ -7.46 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = \mathbf{12,204.42} \ \mathbf{-1,221.74} \ x_i + \mathbf{140.17} \ x_i^2 - \ \mathbf{7.46} \ x_i^3$$

•

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{82,300}{9} = 9,144.44$$

year	$\mathbf{x}_i$	$\mathbf{y}_i$	$\hat{y}_i = \beta_0 + \beta_1 \mathbf{x_i} + \beta_2 \mathbf{x_i^2} + \beta_3 \mathbf{x_i^3}$	**SSE** $= (\mathbf{y}_i - \hat{y}_i)^2$	**SSyy** $= (\mathbf{y}_i - \bar{y})^2$
			12204.4	38,251	
1987	0	12400			10,598,642
			11115.4	46,392	
1988	1	10900			3,081,975
			10261.9	68,595	731,975
1989	2	10000			
			8666.9		126,420
1991	4	9500		174,339	
			8307.8	54,323	59,753
1992	5	8900			
			7960.3	94,746	
1993	6	8000			1,309,753
			7579.7	25,702	
1994	7	7800			1,807,531
			7121.1	413	
1995	8	7600			2,385,309
			82300.0	6,227	
1996	9	7200			3,780,864
			12204.4		
$\Sigma =$	**42**	82,300		**508,988**	**23,882,222**

$$R^2_{cubic\ model} = \frac{SS - SSE}{SS} = \frac{\mathbf{23,882,222} - \mathbf{508,988}}{\mathbf{23,882,222}} = 0.9787$$

•

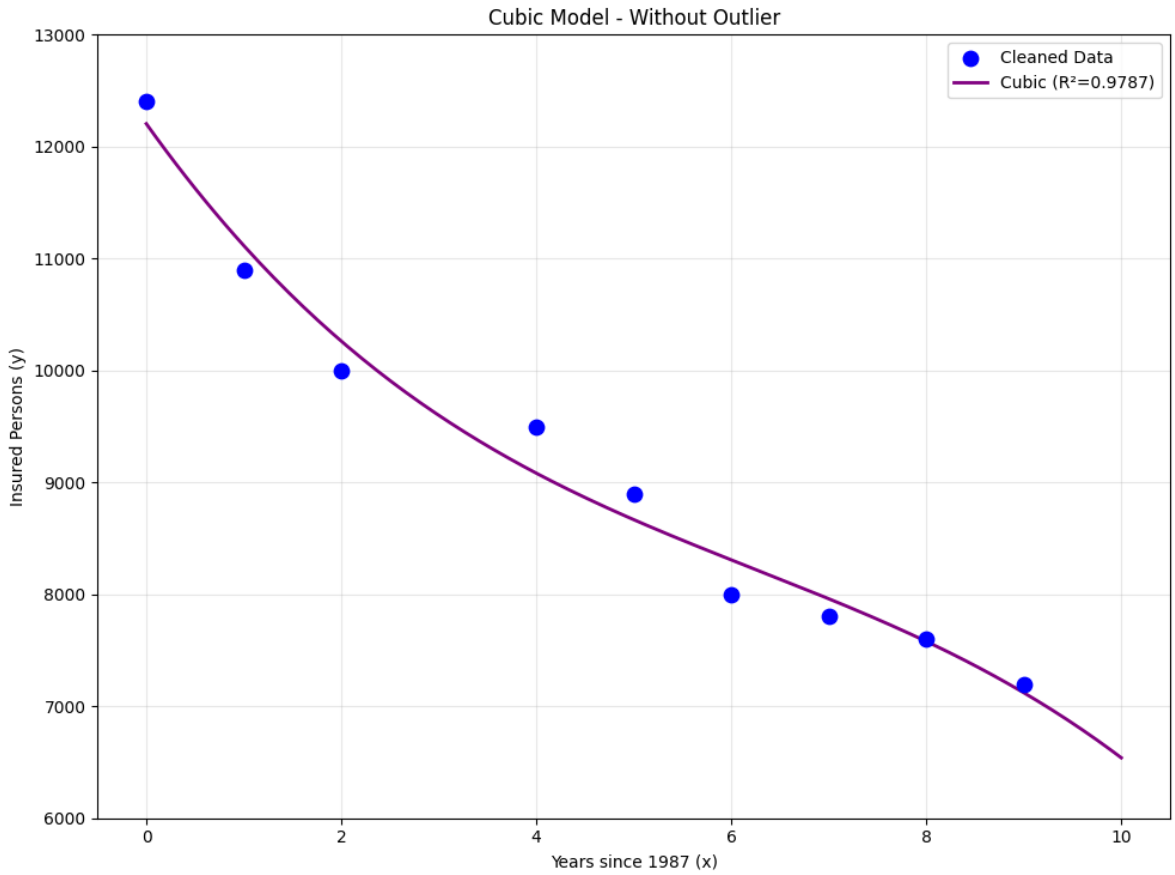


Figure 14: Cubic Model of Cleaned Data without Outlier

•

```
Cubic Model:
  Equation: y = 12204.42 + (-1221.74)*x^1 + (140.17)*x^2 + (-7.46)*x^3
  R^2: 0.9787
```

Figure 15: Simulation Results of Cubic Model of Cleaned Data without Outlier

1.3 Comparison Between Three Models (Cleaned Data):

Model	Equation (MATLAB)	R^2
Linear	$\widehat{y}_i = 11,621.67 \ - 530.83\ x$	0.9439
Quadratic	$\widehat{y}_i = 12,024.17 \ - 863.13\ x + 37.44\ x^2$	0.9725
Cubic	$\widehat{y}_i = 12,204.42 \ - 1,221.74\ x_i + 140.17\ x_i^2 - \ 7.46\ x_i^3$	0.9787

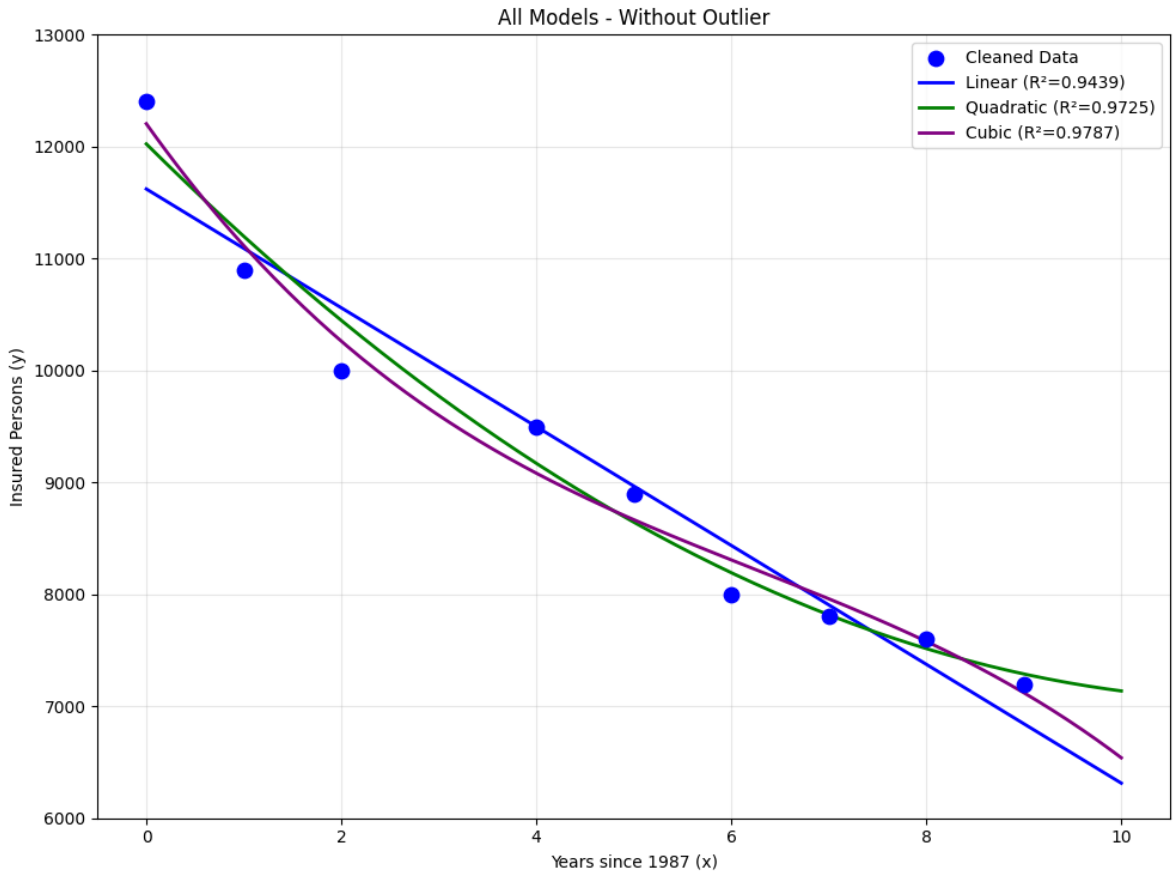


Figure 16: Comparison between three Models of Cleamed Data without Outlier

Comment:

All three R^2 values are very high (> 0.94) after removing the outlier, confirming that it was the primary source of distortion. The linear model alone explains 94.4% of the variance.

1.4 Best-Fit Model Selection

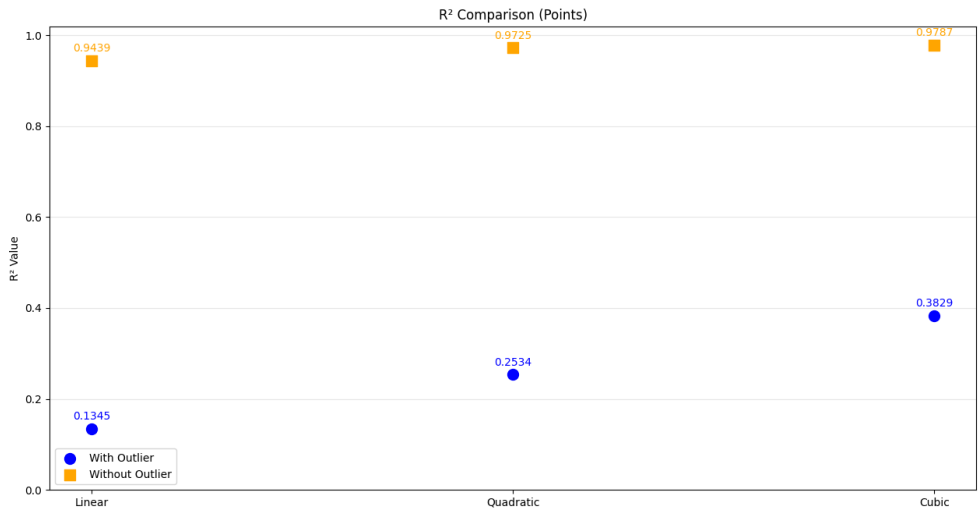


Figure 17: R-square Comparison between three Models

R^2 Comparison

between three Models

Model	Parameters	R^2 (Original) → R^2 (Cleaned)	Gain in R^2 (Original) → Gain in R^2 (Cleaned)
Linear	2	0.1345 → 0.9439	—
Quadratic	3	0.2534 → 0.9725	+0.1189 → +0.0286
Cubic	4	0.3829 → 0.9787	+0.1295 → +0.0062

Conclusion (with outlier):

- Linear → Quadratic: R^2 gain = 0.1189 (significant) → Quadratic is better.
- Quadratic → Cubic: R^2 gain = 0.1295 (significant) → Cubic is better.

Best-fit model (with outlier): Cubic $\hat{y} = 13113.15 - 4313.93\,x + 932.31\,x^2 - 58.95\,x^3$ ($R^2 = 0.3829$)

Conclusion (without outlier):

- Linear → Quadratic: R^2 gain = 0.0286 (significant) → Quadratic is better.
- Quadratic → Cubic: R^2 gain = 0.0062 → gain is NOT significant enough.

Best-fit model (without outlier): Quadratic $\hat{y} = 12{,}024.17 - 863.13\,x + 37.44\,x^2$ ($R^2 = 0.972$)

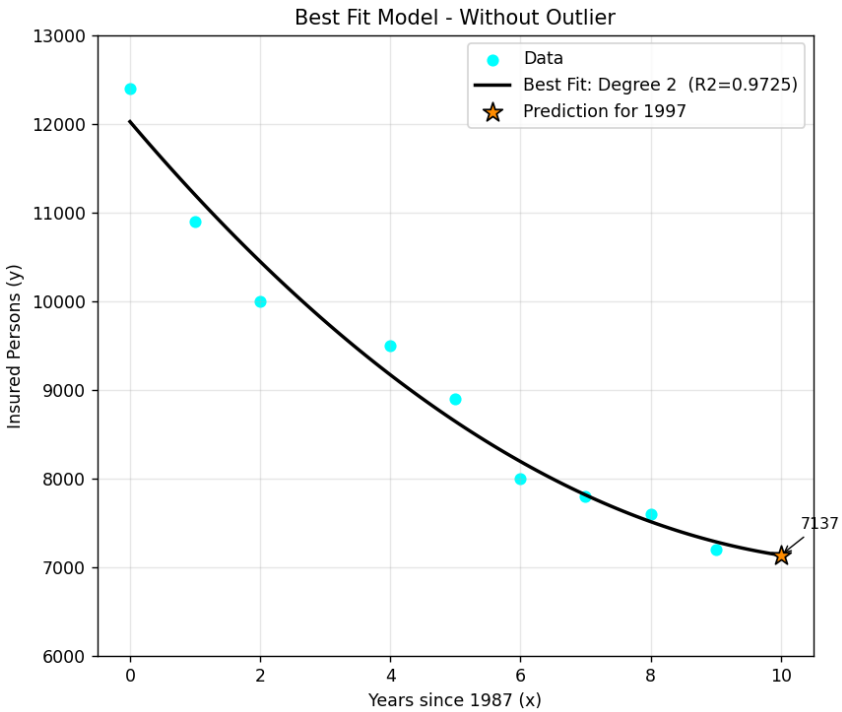


Figure 18: Best Fit Model without Outlier

```
>>  Model Selection: Cubic recommended
   dR2 (Linear -> Quadratic) = +0.1188
   dR2 (Quadratic -> Cubic)  = +0.1295
```

Figure 19: Simulation Results of Best Fit Model with Outlier

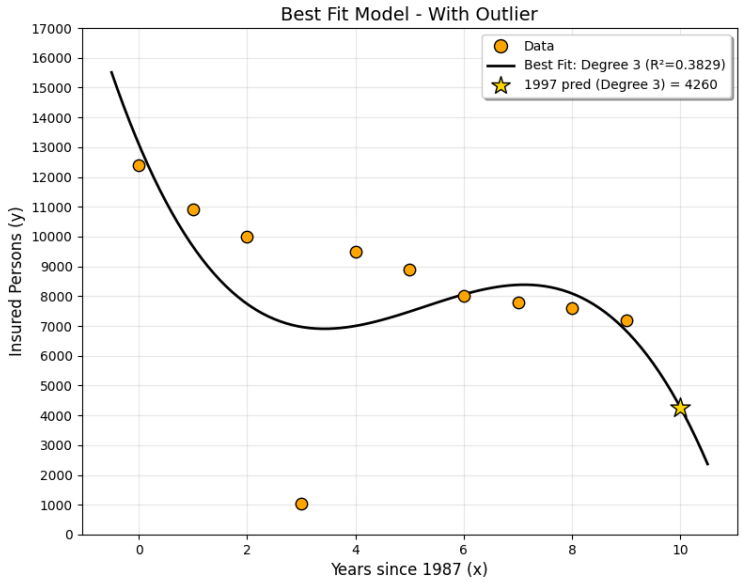


Figure 20: Best Fit Model with Outlier

```
>>  Model Selection: Quadratic recommended
   dR2 (Linear -> Quadratic) = +0.0286
   dR2 (Quadratic -> Cubic)  = +0.0062
```

Figure 21: SImulation Results of Best Fit Model without Outlier

1.5 Prediction of Insured People

At year 1997 (x=10) :

- Using the best-fit model from the original data with outlier (Cubic model as selected):

$$\widehat{y_i} = 13113.15 - 4313.93\ x_i + 932.31\ x_i^2 - \ 58.95\ x_i^3$$

$\hat{y} = 13113.15 - 4313.93(10) + 932.31(10^2) - 58.95(10^3) = 4{,}254.85$

Insured People ≈ 4,254.85 insured person

- Using the best-fit model from the cleaned data without outlier (Quadratic model as selected):

$$\widehat{y_i} = 12,024.17\ - 863.13\ x + 37.44\ x^2$$

$\hat{y} = 12{,}024.17 - 863.13(10) + 37.44(10^2) = 7{,}136.87$

Insured People ≈ 7,136.87 insured person

```
============================================================
  PREDICTIONS FOR 1997  (year = 1997 , x = 10)
============================================================
  Original Data with-outlier (Cubic Model) -> 4,260.00 insured persons

  Cleaned Data without-outlier (Quadratic Model) -> 7,137.06 insured persons
```

Figure 22: Simulation Results of Predection Insured People at year 1997

Predection Insured People at year 1997

Problem 2: Heating Oil

We develop a regression model to estimate heating oil usage for a single-family home in January, based on two predictors: average temperature (Temp, in °F) and amount of insulation (Insulation, in inches).

2.1 Dataset

Oil (gallons)	Temp (°F)	Insulation (in)
270	40	4
362	27	4
162	40	10
45	73	6
91	65	7
233	**65**	**40 ← OUTLIER**
372	10	6
305	9	10
234	24	10
122	65	4
25	66	10
210	41	6
450	22	4
325	40	4
52	60	10

2.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification A 3D scatter plot of Oil Usage vs. Temperature and Insulation reveals that one data point is far outside the cluster formed by the other observations: the point (Temp=65°F, Insulation=40 in, Oil=233 gallons). All other insulation values range from 4 to 10 inches; an insulation thickness of 40 inches is physically unreasonable for a typical single-family home and is almost certainly a data entry error (e.g., the value 4 recorded as 40). This point is identified as the outlier and is highlighted in red in the scatter plot below.

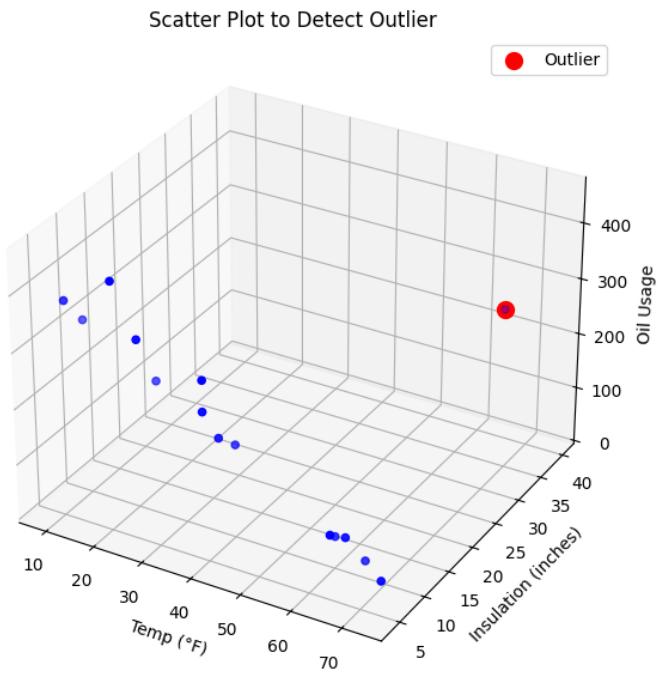


Figure 23: 3D Scatter Plot to Detect Outlier (outlier in red: Insulation=40)

Detect Outlier (outlier in red: Insulation=40)

2. Regression Models

2.1 Linear Model

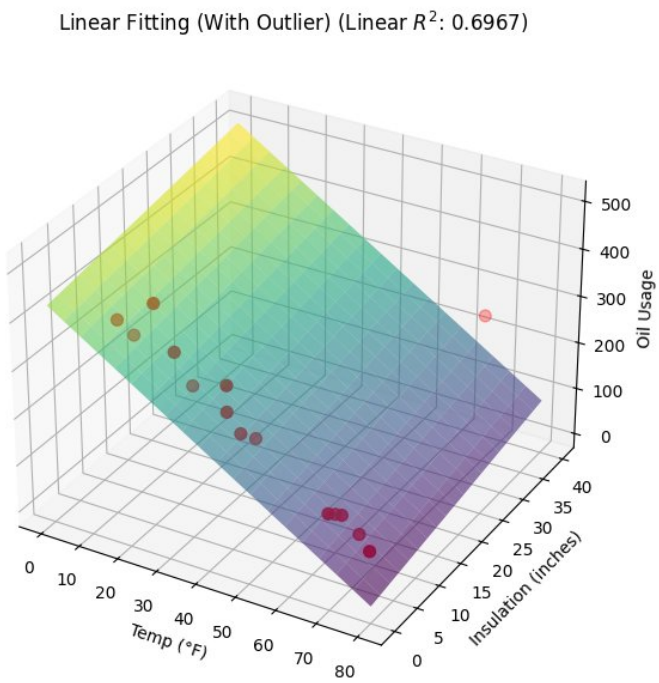


Figure 24: 3D Linear Regression Surface (With Outlier), R2 = 0.6967

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1 * \ T \ + \beta_2 * \ I$$

where T = Temp, I = Insulation

$$Oil = 426.97 \ - \ 5.274 * T \ + 1.966 * I$$

```
Linear model:
  Intercept : 426.967656
  Coef      : Temp=-5.273465,  Insulation=1.966053
  Equation  : y = 426.9677  +  (-5.2735)*Temp  +  (1.9661)*Insulation
  R2        : 0.696686
```

Figure 25: Simulation Results of Linear Regression Surface (With Outlier)

So, the linear fit equation for the original dataset is:

$$\widehat{y_i} = 426.97 - \ 5.27x_{1i} + 1.966 \ \ x_{2i}$$

The Coefficient of Determination (R^2):

R^2 (Linear, With Outlier) = 0.6967

2.2 Quadratic Model

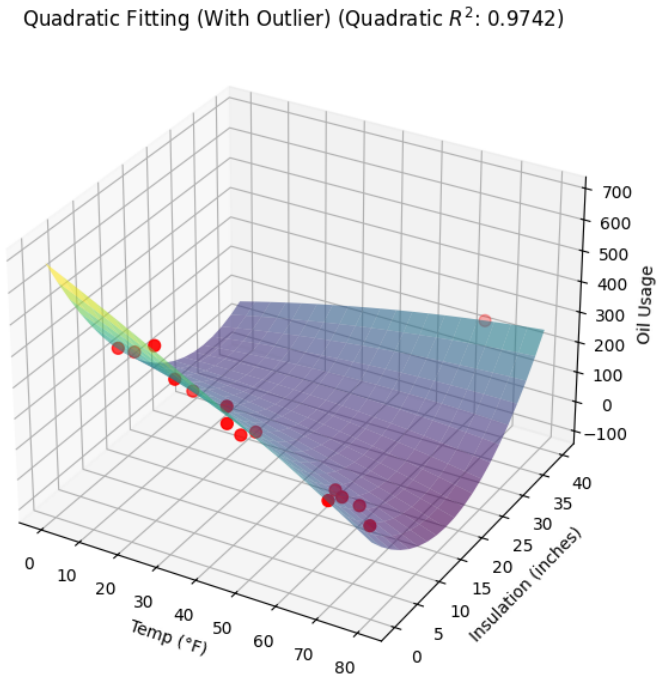


Figure 26: 3D Quadratic Regression Surface (With Outlier), R2 = 0.9742

Surface (With Outlier), $R^2 = 0.9742$

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1\ T\ + \beta_2\ I\ + \beta_3 T^2 + \beta_4 TI + \beta_5\ I^2$$

```
Quadratic model:
  Intercept : 680.025958
  Temp                : -6.167784
  Insulation          : -41.716060
  Temp^2              : -0.012475
  Temp Insulation     : 0.239147
  Insulation^2        : 0.658593
  Equation  : y = 680.0260  +  (-6.1678)*Temp  +  (-41.7161)*Insulation  +  (-0.0125)*Temp^2  +  (0.2391)*Temp
  Insulation  +  (0.6586)*Insulation^2
  R2        : 0.974152
```

Figure 27: Simulation Results of Quadratic Regression Surface (With Outlier)

where T = Temp, I = Insulation

$$Oil = 680.03\ -\ 6.168\ T\ -\ 41.716\ I\ -\ 0.01248\ T^2 + 0.2392\ TI + 0.6586\ I^2$$

So, the quadratic fit equation for the original dataset is:

$$\widehat{y_i} = 680\ -\ 6.17x_{1i} - 41.72\ x_{2i} -\ 0.012\ x_{1i}^2 + 0.24\ x_{1i}x_{2i} + 0.66\ x_{2i}^2$$

The Coefficient of Determination (R^2):

R^2 (Quadratic, With Outlier) = 0.9742

2.3 Case B: WITHOUT Outlier

1. Scatter Plot & Outlier Identification (WITHOUT Outlier) The data point (Temp=65, Insulation=40, Oil=233) is removed. All remaining 14 data points have insulation values between 4 and 10 inches, which is physically reasonable. The cleaned scatter plot is shown below.

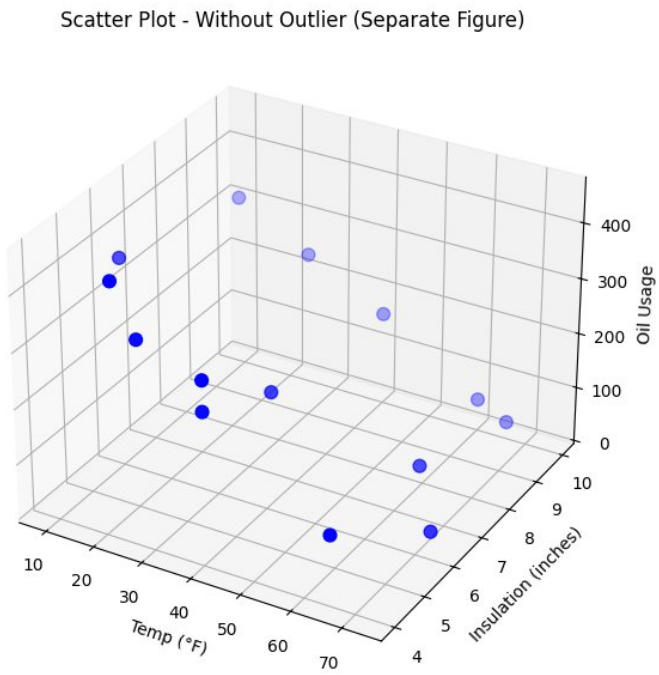


Figure 28: 3D Scatter Plot (Cleaned Data, Outlier Removed)

2. Regression Models (WITHOUT Outlier)

2.1 Linear Model

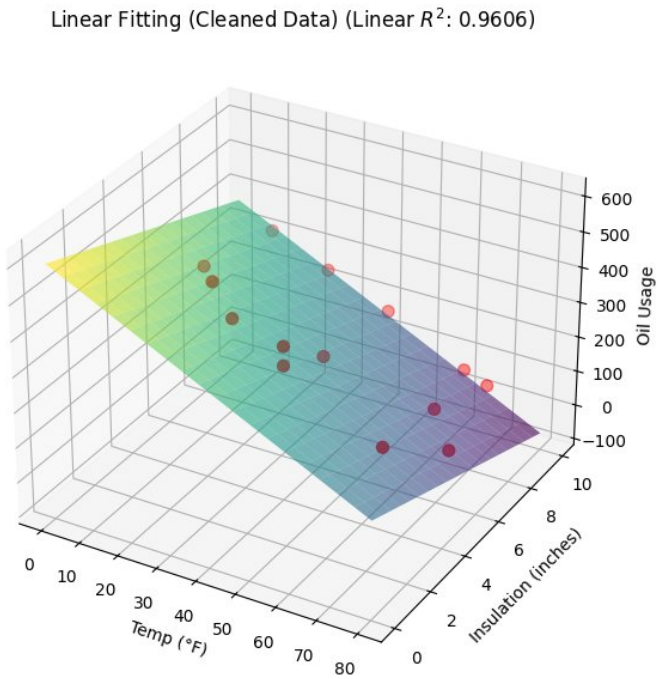


Figure 29: 3D Linear Regression Surface (Cleaned Data), R2 = 0.9606

Surface (Cleaned Data), $R^2 = 0.9606$

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1 * \ T \ + \beta_2 * \ I$$

where T = Temp, I = Insulation

$$Oil = 601.6 \ - \ 5.482 \ T \ - \ 23.232 \ I$$

```
Linear model:
  Intercept : 601.601412
  Coef      : Temp=-5.481747,  Insulation=-23.232031
  Equation  : y = 601.6014  +  (-5.4817)*Temp  +  (-23.2320)*Insulation
  R2        : 0.960609
```

Figure 30: Simulation Results of Linear Regression Surface (Cleaned Data)

So, the linear fit equation for the original dataset is:

$$\widehat{y_i} = 601.6 - \ 5.482 \ x_{1i} + 23.232 \ \ x_{2i}$$

The Coefficient of Determination (R^2):
R^2 (Linear, With Outlier) = 0.96061

2.2 Quadratic Model

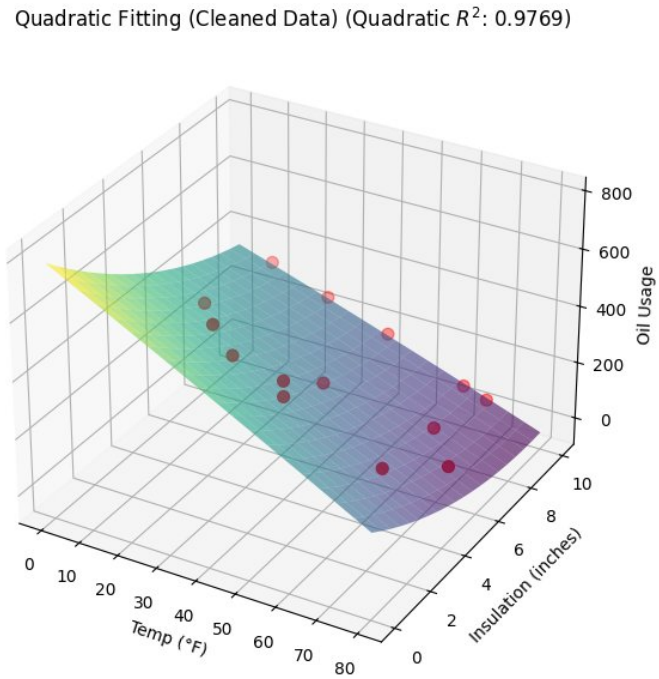


Figure 31: 3D Quadratic Regression Surface (Cleaned Data), R^2 = 0.9769

Surface (Cleaned Data), R^2 = 0.9769

The multivariate quadratic model has the form:

$$Oil = \beta_0 + \beta_1 \ T \ + \beta_2 \ I \ + \beta_3 T^2 + \beta_4 TI + \beta_5 \ I^2$$

where T = Temp, I = Insulation

$$Oil = 785.52 - \ 6.963 \ T \ - \ 72.159 \ I \ - \ 0.00348 \ T^2 + 0.2567 \ TI + 2.7438 \ I^2$$

```
Quadratic model:
  Intercept : 785.516872
  Temp                : -6.962462
  Insulation          : -72.158742
  Temp^2              : -0.003478
  Temp Insulation     : 0.256701
  Insulation^2        : 2.743800
  Equation  : y = 785.5169  +  (-6.9625)*Temp  +  (-72.1587)*Insulation  +  (-0.0035)*Temp^2  +  (0.2567)*Temp
  Insulation  +  (2.7438)*Insulation^2
  R2        : 0.976889
```

Figure 32: Simulation Results of Quadratic Regression Surface (Cleaned Data)

So, the linear fit equation for the original dataset is:

$$\widehat{y}_i = 785.52 \; - \; 6.963 \; x_{1i} - 72.159 \; x_{2i} - \; 0.00348 \; x_{1i}^2 + 0.2567 \; x_{1i}x_{2i} + 2.7438 \; x_{2i}^2$$

The Coefficient of Determination (R^2):

R^2 (Quadratic, With Outlier) = 0.9769

2.4 Comparison of Models and Best-Fit Selection

Model	R^2 (With Outlier)	$R^{2\ 2}$ (Without Outlier)	Gain (With Outlier)	Gain ^{2} (Without Outlier)
Linear	0.6967	0.9606	—	—
Quadratic	0.9742	0.9769	+0.2775	+0.0163

Conclusion (Part a — With Outlier):

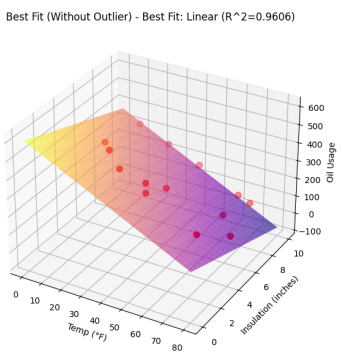
Linear → Quadratic gain = +0.2775

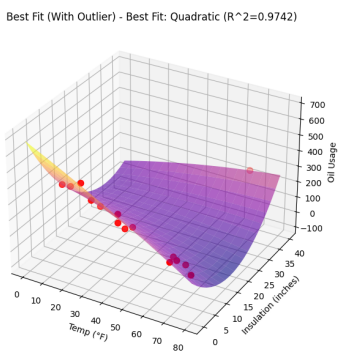
Best-fit model (with outlier): Quadratic (R^2 = 0.9742).

Conclusion (Part b — Without Data):

Linear → Quadratic gain = +0.0163 (NOT significant enough).

Best-fit model (without outlier): Linear (R^2 = 0.9606).





After removing the outlier, the linear model already achieves R^2 = 0.9606. The gain from linear to quadratic is only +0.0163, which is NOT significant enough as specified in Note 4 of the assignment.

- **Simulation Results:**

```
Part a  (With Outlier):
  dR2 (Linear -> Quadratic) = +0.2775  -->  significant gain
  Best-fit model: Quadratic  (R2 = 0.9742)

Part b  (Without Outlier):
  dR2 (Linear -> Quadratic) = +0.0163  -->  not significant
  Best-fit model: Linear     (R2 = 0.9606)
```

Figure 33: Simulation Results of Best Fit Model

2.5 Prediction of Oil

Temp =15°F, Insulation = 5 inches

The linear model could be abstracted as follows:

$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$

Linear Model	**Data with Outlier**	**Data without Outlier (Best Fit)**
β_0	427	601.6
β_1	- 5.273	- 5.482
β_2	1.966	- 23.23
$\mathbf{R}^2$	0.6966	0.9606
$\hat{y}$**(temp = 15, insulation = 5)**	357.7	403.22

- Using the best-fit model from the cleaned data without outlier (Linear model as selected):

Oil = 601.60 − 5.482 * 15 − 23.232 * 5 = 403.21

Predicted Oil Usage ≈ 403.2 gallons

- **Simulation Results:**

```
Prediction at (15F,5in) using CLEANED Linear best-fit: 403.22 units
```

Figure 34: Simulation Results of Prediction Oil of Cleaned Data

Prediction Oil of Cleaned Data

The quadratic model could be abstracted as follows:

$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 {x_1}^2{}_i + \beta_4 x_{1i} x_{2i} + \beta_5 {x_2}^2{}_i$

Linear Model	**Data with Outlier (Best Fit)**	**Data without Outlier**
β_0	680	785.5
β_1	- 6.17	- 6.96
β_2	- 41.72	- 72.16
β_3	- 0.012	- 0.0034
β_4	0.24	0.2567
β_5	0.66	2.744
$\mathbf{R}^2$	0.9741	0.9768
$\hat{y}$**(temp = 15, insulation = 5)**	410.52	407.35

- Using the best-fit model from the original data with outlier (Quadratic model as selected):

Oil = 680 + (−6.17)(15) + (−41.72)(5) + (−0.012)(15^2) + (0.24)(15)(5) + (0.66)(5^2) = 410.65

Predicted Oil Usage ≈ 410.65 gallons

- **Simulation Results:**

```
Prediction at (15F,5in) using ORIGINAL Quadratic best-fit: 410.52 units
```

Figure 35: Simulation Results of Prediction Oil of Original Data

Prediction Oil of Original Data